SLM PLATFORM
PRODUCT GUIDE

THE COMPLETE STRATEGY, IMPLEMENTATION,
AND ADVANCED ANALYTICS HANDBOOK

SLM Platform: End-User Step-by-Step Guide

Version 1.0 -- Production-Ready Fine-Tuning & Evaluation

Welcome to the SLM Platform. This guide provides step-by-step instructions on how to use the web interface to take raw data, process it, train a Small Language Model (SLM), and evaluate its exact performance via our LLM-as-a-Judge benchmarking suite.

1. Project Setup and Data Ingestion

The first step is bringing your raw documents into the platform so they can be processed into training data.

1. **Access the Dashboard**: Open your web browser and navigate to the platform UI (e.g., `http://localhost:5173`).
2. **Create a Project**: Click on **Create New Project**. Give it a descriptive name like "Customer Support Helper" and set a description.
3. **Navigate to Data Ingestion**: Click into your new project, and select the **Data Ingestion** tab on the left sidebar.
4. **Upload Files**: Drag and drop your source documents (PDFs, Markdown, TXT, or CSVs) into the upload zone. Click **Upload and Process**. The backend will extract the text and split it into manageable chunks.

2. Data Cleaning and Preparation

Raw text usually contains noise and sensitive information. The platform automates the sanitization process.

1. **Go to Data Cleaning**: Click the **Data Cleaning** tab.
2. **Select Deduplication**: Check the box for exact or semantic deduplication to remove repeated text blocks from your dataset.
3. **PII Redaction**: Check the box for PII redaction if your data contains sensitive user information (like emails or SSNs).
4. **Run Cleaning**: Click **Start Cleaning Job**. The progress bar will update as the backend sanitizes your text.

3. Building the Gold Dataset & Synthetic Generation

To train an SLM, you need high-quality Question & Answer (Q&A) pairs.

1. **Manual Gold Set Creation**: Go to the **Gold DatasetBuilder** tab. Here, you can manually type in perfect Q&A pairs that represent the exact behavior you want the model to learn. Mark these with high difficulty/criticality.
2. **Synthetic Data Generation**: If manual creation is too slow, go to the **Synthetic Data Generator** tab.
3. **Select a Teacher Model**: Choose a large, highly capable model (e.g., Llama-3-70B) from the dropdown.
4. **Generate Pairs**: Click **Generate Q&A Pairs**. The Teacher model will read your ingested documents and automatically generate diverse questions and answers based on the text.
5. **Review and Approve**: Review the generated pairs and click "Approve" for the good ones. These approved pairs are saved to your dataset.

4. Tokenization

Before training, the text must be converted into tokens that the model understands.

1. **Go to Tokenization**: Select the **Tokenization** tab.
2. **Select Base Model**: Choose the base SLM you intend to train (e.g., `meta-llama/Meta-Llama-3-8B`). The appropriate tokenizer will be loaded automatically.
3. **Run Tokenization**: Click **Start Tokenization**. The system will format all approved Q&A pairs into the correct chat template (like ChatML or Llama-3 instruction format) and pack the sequences for GPU efficiency.

5. Training the SLM (Fine-Tuning)

This is where the model actually learns from your data.

1. **Navigate to Training**: Click the **Training Dashboard** tab.
2. **Create New Experiment**: Click **New Experiment** to set up a training run.
3. **Configure Settings**:
4. **Base Model**: Put the HuggingFace ID of your starting model.
5. **LoRA Configurations**: Adjust Rank (r), Alpha, and target modules depending on how deeply you want to fine-tune the model.
6. **Optimizer**: Select **Paged AdamW 8-bit** to save VRAM on your DGX Spark, or standard AdamW if memory allows.
7. **Start Training**: Click **Launch Training Job**.
8. **Monitor Live Telemetry**: A real-time dashboard powered by WebSockets will appear. You will see live charts updating with the model's **Training Loss**, **Learning Rate Decay**, and simulated **GPU Utilization**. Wait for the loss curve to flatten and the job to complete.

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6. Evaluation and LLM-as-a-Judge Benchmarking

Once trained, you must verify the model's quality against ground truth.

1. **Go to Evaluation**: Click the **Evaluation** tab.
2. **Select the Trained Experiment**: Click on the completed "Test Run" experiment to load its context.
3. **Run LLM Benchmark**: Click the **+ Run LLM Benchmark** button.
4. **Configure Judge**: In the form, verify the Judge Model (e.g., `meta-llama/Meta-Llama-3-70B-Instruct`) and select the benchmark suite (e.g., MMLU-Subset). Click **Run Benchmark**.
5. **Review Radar Chart**: Once complete, a **Skill Radar Chart** will appear, visualizing your model's pass rate across different domains (Logic, Tone, Factuality, etc.).
6. **Side-by-Side Arena**: Scroll down to the **LLM Judge: Side-by-Side Comparison**. Here you can see:
7. The original Prompt.
8. The Ground Truth Reference answer.
9. Your newly trained SLM's Prediction.
10. The **Judge Score (Out of 5)** and the technical **Rationale** provided by the Teacher model evaluating the answer.

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7. Model Export and Compression

When you are satisfied with the evaluation results, prepare the model for deployment.

1. **Go to Export & Compression**: Navigate to the **Export** tab.
2. **Quantization Options**: Choose if you want to compress the weights to 8-bit or 4-bit to make the model run faster on edge devices.
3. **Format Selection**: Choose the output format (e.g., GGUF for local inference, or ONNX).
4. **Export**: Click **Export Model**. The platform will package your fine-tuned weights, merge them with the base model, and generate the final deployment-ready files.